

What should you recall about the particle model?

CHE-0001

PREGUNTA

What should you recall about changes of state?

CHE-0002

PREGUNTA

What exam pitfall should you avoid in diffusion?

CHE-0003

PREGUNTA

Which statement describes particles in a liquid?
A) They vibrate only at fixed positions.
B) They are close and move past one another.
C) They are very far apart and stationary.
D) They form a regular lattice and cannot move.

CHE-0004

PREGUNTA

Which process occurs throughout a liquid at a fixed temperature?
A) evaporation
B) sublimation
C) boiling
D) diffusion

CHE-0005

PREGUNTA

Why does gas pressure increase when a sealed gas is heated?
A) Particles become larger.
B) Particles collide more often and harder.
C) The number of particles increases.
D) Attractive forces become stronger.

CHE-0006

PREGUNTA

Which change is exothermic?
A) melting
B) boiling
C) sublimation
D) freezing

CHE-0007

PREGUNTA

What should you recall about elements, compounds and mixtures?

CHE-0008

PREGUNTA

What should you recall about atomic structure?

CHE-0009

PREGUNTA

State the key equation or relationship for atomic structure.

CHE-0010

PREGUNTA

Which statement defines a compound?
A) One type of atom only
B) Elements chemically combined in fixed proportions
C) Substances mixed in any proportion
D) A substance separable by filtration

CHE-0011

PREGUNTA

An atom has proton number 12 and nucleon number 25. How many neutrons?
A) 12
B) 13
C) 25
D) 37

CHE-0012

PREGUNTA

FagusED · Chemistry
01 · States of matter

Use relative molecular mass to compare gaseous diffusion rates; do not attribute diffusion to gravity.

CHE-0003

RESPUESTA

FagusED · Chemistry
01 · States of matter

Melting and boiling require energy to overcome attractive forces between particles. Freezing and condensation release energy as stronger attractions form. Temperature remains constant during a pure substance's change of state because supplied energy changes potential energy rather than average kinetic energy.

CHE-0002

RESPUESTA

FagusED · Chemistry
01 · States of matter

All matter is made of particles in constant motion. In a solid, particles are closely packed in a regular arrangement and vibrate about fixed positions. In a liquid, particles remain close but move past one another. In a gas, particles are far apart, move rapidly and randomly, and collide with the container.

CHE-0001

RESPUESTA

FagusED · Chemistry
01 · States of matter

B) Particles collide more often and harder. — Higher kinetic energy produces more frequent, larger-force collisions.

CHE-0006

RESPUESTA

FagusED · Chemistry
01 · States of matter

C) boiling — Boiling occurs throughout the liquid at its boiling point.

CHE-0005

RESPUESTA

FagusED · Chemistry
01 · States of matter

B) They are close and move past one another. — Liquid particles remain close but can change neighbours.

CHE-0004

RESPUESTA

FagusED · Chemistry
02 · Atoms, elements and compounds

Protons and neutrons occupy the nucleus; electrons occupy shells. Proton number identifies the element. Nucleon number is protons plus neutrons. In a neutral atom, electrons equal protons. The first shell holds 2 electrons and subsequent syllabus-level arrangements are written, for example, 2,8,7.

CHE-0009

RESPUESTA

FagusED · Chemistry
02 · Atoms, elements and compounds

An element contains one type of atom. A compound contains two or more elements chemically combined in fixed proportions and can be separated only chemically. A mixture contains substances not chemically combined; its components retain their properties and can be separated physically.

CHE-0008

RESPUESTA

FagusED · Chemistry
01 · States of matter

D) freezing — Freezing releases energy as particles form stronger attractions.

CHE-0007

RESPUESTA

FagusED · Chemistry
02 · Atoms, elements and compounds

B) $13 - 25 - 12 = 13$.

CHE-0012

RESPUESTA

FagusED · Chemistry
02 · Atoms, elements and compounds

B) Elements chemically combined in fixed proportions — A compound contains chemically bonded elements in fixed ratios.

CHE-0011

RESPUESTA

FagusED · Chemistry
02 · Atoms, elements and compounds

neutrons = nucleon number - proton number

CHE-0010

RESPUESTA

FagusED · Chemistry
02 · Atoms, elements and compounds

Which particles determine an element's chemical properties?

- A) nuclear neutrons
- B) outer-shell electrons
- C) all nucleons
- D) only inner electrons

CHE-0013

PREGUNTA

FagusED · Chemistry
02 · Atoms, elements and compounds

Isotopes of one element have different numbers of

- A) protons
- B) outer shells
- C) neutrons
- D) nuclear charges

CHE-0014

PREGUNTA

FagusED · Chemistry
03 · Stoichiometry

What should you recall about formulae and equations?

CHE-0015

PREGUNTA

FagusED · Chemistry
03 · Stoichiometry

What should you recall about relative masses and the mole?

CHE-0016

PREGUNTA

FagusED · Chemistry
03 · Stoichiometry

State the key equation or relationship for relative masses and the mole.

CHE-0017

PREGUNTA

FagusED · Chemistry
03 · Stoichiometry

What is the M_r of CaCO_3 ? Ar: Ca 40, C 12, O 16.

- A) 68
- B) 84
- C) 100
- D) 116

CHE-0018

PREGUNTA

FagusED · Chemistry
03 · Stoichiometry

How many moles are in 11 g of CO_2 ?

- A) 0.25
- B) 0.50
- C) 2.0
- D) 4.0

CHE-0019

PREGUNTA

FagusED · Chemistry
03 · Stoichiometry

Which is a correct balanced equation?

- A) $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$
- B) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- C) $\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$
- D) $2\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$

CHE-0020

PREGUNTA

FagusED · Chemistry
03 · Stoichiometry

How many particles are in 2.0 mol?

- A) 3.01×10^{23}
- B) 6.02×10^{23}
- C) 1.20×10^{24}
- D) 2.00×10^{24}

CHE-0021

PREGUNTA

FagusED · Chemistry
04 · Electrochemistry

What should you recall about electrolysis and electrodes?

CHE-0022

PREGUNTA

FagusED · Chemistry
04 · Electrochemistry

What should you recall about molten ionic compounds?

CHE-0023

PREGUNTA

FagusED · Chemistry
04 · Electrochemistry

State the key equation or relationship for electrolysis and electrodes.

CHE-0024

PREGUNTA

FagusED · Chemistry
04 · Electrochemistry

Where does reduction occur during electrolysis?

- A) anode
- B) cathode
- C) electrolyte surface
- D) power supply only

CHE-0025

PREGUNTA

FagusED · Chemistry
04 · Electrochemistry

Why does solid sodium chloride not conduct?

- A) It has no ions.
- B) Its ions are fixed.
- C) Its electrons are delocalised.
- D) Sodium is absent.

CHE-0026

PREGUNTA

FagusED · Chemistry
04 · Electrochemistry

What forms at the cathode during electrolysis of molten PbBr_2 ?

- A) bromine
- B) oxygen
- C) lead
- D) hydrogen

CHE-0027

PREGUNTA

FagusED · Chemistry
03 · Stoichiometry

A formula shows the simplest whole-number ratio of atoms or ions. Balanced equations conserve the number of each type of atom and therefore mass. State symbols are (s), (l), (g) and (aq). Ionic equations remove spectator ions and show only species that change.

CHE-0015

RESPUESTA

FagusED · Chemistry
02 · Atoms, elements and compounds

C) neutrons — They share proton number but differ in neutrons.

CHE-0014

RESPUESTA

FagusED · Chemistry
02 · Atoms, elements and compounds

B) outer-shell electrons — Bonding and reactions involve valence electrons.

CHE-0013

RESPUESTA

FagusED · Chemistry
03 · Stoichiometry

C) $100 - 40 + 12 + 3 \times 16 = 100$.

CHE-0018

RESPUESTA

FagusED · Chemistry
03 · Stoichiometry

$n = m/M$ $N = nL$

CHE-0017

RESPUESTA

FagusED · Chemistry
03 · Stoichiometry

Relative molecular/formula mass, M_r , is the sum of A_r values. One mole contains the Avogadro constant, 6.02×10^{23} , of specified particles. Molar mass has units g/mol.

CHE-0016

RESPUESTA

FagusED · Chemistry
03 · Stoichiometry

C) $1.20 \times 10^{24} - 2.0 \times 6.02 \times 10^{23}$.

CHE-0021

RESPUESTA

FagusED · Chemistry
03 · Stoichiometry

B) $2H_2 + O_2 \rightarrow 2H_2O$ — It conserves four H and two O atoms.

CHE-0020

RESPUESTA

FagusED · Chemistry
03 · Stoichiometry

A) 0.25 — M_r 44, so $11/44 = 0.25$ mol.

CHE-0019

RESPUESTA

FagusED · Chemistry
04 · Electrochemistry

cathode: reduction anode: oxidation

CHE-0024

RESPUESTA

FagusED · Chemistry
04 · Electrochemistry

A molten compound contains only its own ions. At the cathode the metal cation gains electrons; at the anode the non-metal anion loses electrons. For molten lead(II) bromide, Pb^{2+} forms lead and Br^- forms bromine.

CHE-0023

RESPUESTA

FagusED · Chemistry
04 · Electrochemistry

Electrolysis is the decomposition of an ionic compound, molten or in aqueous solution, by electricity. The electrolyte contains mobile ions. The cathode is negative and attracts cations; reduction (gain of electrons) occurs there. The anode is positive and attracts anions; oxidation (loss of electrons) occurs there.

CHE-0022

RESPUESTA

FagusED · Chemistry
04 · Electrochemistry

C) lead — Pb^{2+} ions gain electrons.

CHE-0027

RESPUESTA

FagusED · Chemistry
04 · Electrochemistry

B) Its ions are fixed. — The ions cannot carry charge through the solid.

CHE-0026

RESPUESTA

FagusED · Chemistry
04 · Electrochemistry

B) cathode — Cations gain electrons at the cathode.

CHE-0025

RESPUESTA

FagusED · Chemistry
04 · Electrochemistry

Which half-equation is oxidation?

- A) $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$
- B) $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
- C) $\text{H}^+ + \text{e}^- \rightarrow \text{H}$
- D) $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$

CHE-0028

PREGUNTA

FagusED · Chemistry
05 · Chemical energetics

What should you recall about energy transfer?

CHE-0029

PREGUNTA

FagusED · Chemistry
05 · Chemical energetics

What should you recall about reaction pathway diagrams?

CHE-0030

PREGUNTA

FagusED · Chemistry
05 · Chemical energetics

State the key equation or relationship for energy transfer.

CHE-0031

PREGUNTA

FagusED · Chemistry
05 · Chemical energetics

Which change is exothermic?

- A) photosynthesis
- B) thermal decomposition
- C) combustion
- D) bond breaking only

CHE-0032

PREGUNTA

FagusED · Chemistry
05 · Chemical energetics

In an endothermic reaction, products have

- A) less energy than reactants
- B) more energy than reactants
- C) zero energy
- D) the same energy always

CHE-0033

PREGUNTA

FagusED · Chemistry
05 · Chemical energetics

Activation energy is the

- A) total energy in products
- B) minimum energy for successful collision
- C) energy released by cooling
- D) temperature change

CHE-0034

PREGUNTA

FagusED · Chemistry
05 · Chemical energetics

A catalyst changes the

- A) overall enthalpy change
- B) product energy
- C) activation energy
- D) reactant formula

CHE-0035

PREGUNTA

FagusED · Chemistry
06 · Chemical reactions

What should you recall about physical and chemical changes?

CHE-0036

PREGUNTA

FagusED · Chemistry
06 · Chemical reactions

What should you recall about collision theory and rate?

CHE-0037

PREGUNTA

FagusED · Chemistry
06 · Chemical reactions

State the key equation or relationship for collision theory and rate.

CHE-0038

PREGUNTA

FagusED · Chemistry
06 · Chemical reactions

Which is a chemical change?

- A) melting ice
- B) boiling water
- C) burning magnesium
- D) dissolving sugar

CHE-0039

PREGUNTA

FagusED · Chemistry
05 · Chemical energetics

A pathway diagram plots energy against progress of reaction. Activation energy is the minimum energy colliding particles need for a successful reaction. The energy change is product energy minus reactant energy. A catalyst provides an alternative pathway with lower activation energy but does not change reactant energy, product energy or overall energy change.

CHE-0030

RESPUESTA

FagusED · Chemistry
05 · Chemical energetics

Exothermic reactions transfer thermal energy to the surroundings, causing temperature to rise; products have lower chemical energy than reactants. Endothermic reactions take in thermal energy, causing temperature to fall; products have higher chemical energy. Combustion and neutralisation are exothermic; thermal decomposition and photosynthesis are endothermic.

CHE-0029

RESPUESTA

FagusED · Chemistry
04 · Electrochemistry

B) $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ — Electrons appear as products, so they are lost.

CHE-0028

RESPUESTA

FagusED · Chemistry
05 · Chemical energetics

B) more energy than reactants — The system has absorbed energy.

CHE-0033

RESPUESTA

FagusED · Chemistry
05 · Chemical energetics

C) combustion — Combustion transfers energy to surroundings.

CHE-0032

RESPUESTA

FagusED · Chemistry
05 · Chemical energetics

Delta H 0 endothermic

CHE-0031

RESPUESTA

FagusED · Chemistry
06 · Chemical reactions

A physical change does not form a new substance and is often reversible by physical means. A chemical change rearranges atoms and forms new substances; signs can include gas, precipitate, lasting colour change or temperature change, but evidence should be interpreted in context. Mass is conserved in a closed system.

CHE-0036

RESPUESTA

FagusED · Chemistry
05 · Chemical energetics

C) activation energy — It supplies a lower-energy pathway.

CHE-0035

RESPUESTA

FagusED · Chemistry
05 · Chemical energetics

B) minimum energy for successful collision — It is the threshold for reaction.

CHE-0034

RESPUESTA

FagusED · Chemistry
06 · Chemical reactions

C) burning magnesium — Magnesium oxide, a new substance, forms.

CHE-0039

RESPUESTA

FagusED · Chemistry
06 · Chemical reactions

rate = quantity of reactant used or product formed / time

CHE-0038

RESPUESTA

FagusED · Chemistry
06 · Chemical reactions

Particles react only when they collide with sufficient energy and suitable orientation. Rate increases with concentration or gas pressure because collisions are more frequent; with surface area because more particles are exposed; and with temperature because collisions are more frequent and a larger fraction exceed activation energy.

CHE-0037

RESPUESTA

FagusED · Chemistry
06 · Chemical reactions

Rate can be calculated as

- A) time/change
- B) change/time
- C) mass×time
- D) volume+time

CHE-0040

PREGUNTA

FagusED · Chemistry
06 · Chemical reactions

Why does higher concentration increase rate?

- A) Particles get larger.
- B) More frequent collisions occur.
- C) Activation energy increases.
- D) Bonds vanish.

CHE-0041

PREGUNTA

FagusED · Chemistry
06 · Chemical reactions

Why does higher temperature have a large effect?

- A) Fewer particles move.
- B) More particles exceed activation energy.
- C) Concentration becomes zero.
- D) Catalyst is formed.

CHE-0042

PREGUNTA

FagusED · Chemistry
07 · Acids, bases and salts

What should you recall about acids, bases and alkalis?

CHE-0043

PREGUNTA

FagusED · Chemistry
07 · Acids, bases and salts

What should you recall about ph and indicators?

CHE-0044

PREGUNTA

FagusED · Chemistry
07 · Acids, bases and salts

State the key equation or relationship for neutralisation and ionic equations.

CHE-0045

PREGUNTA

FagusED · Chemistry
07 · Acids, bases and salts

Which ion causes aqueous acidity?

- A) H+
- B) OH-
- C) Na+
- D) Cl-

CHE-0046

PREGUNTA

FagusED · Chemistry
07 · Acids, bases and salts

Acid + carbonate produces salt, water and

- A) hydrogen
- B) oxygen
- C) carbon dioxide
- D) chlorine

CHE-0047

PREGUNTA

FagusED · Chemistry
07 · Acids, bases and salts

Sulfuric acid forms

- A) chlorides
- B) nitrates
- C) sulfates
- D) carbonates

CHE-0048

PREGUNTA

FagusED · Chemistry
07 · Acids, bases and salts

Which pH is strongly alkaline?

- A) 1
- B) 5
- C) 7
- D) 13

CHE-0049

PREGUNTA

FagusED · Chemistry
08 · The Periodic Table

What should you recall about arrangement and electronic structure?

CHE-0050

PREGUNTA

FagusED · Chemistry
08 · The Periodic Table

What should you recall about group i: alkali metals?

CHE-0051

PREGUNTA

FagusED · Chemistry
06 · Chemical reactions

B) More particles exceed activation energy. — The energetic fraction rises.

CHE-0042

RESPUESTA

FagusED · Chemistry
06 · Chemical reactions

B) More frequent collisions occur.
— More particles per volume collide more often.

CHE-0041

RESPUESTA

FagusED · Chemistry
06 · Chemical reactions

B) change/time — Rate is change in quantity per unit time.

CHE-0040

RESPUESTA

FagusED · Chemistry
07 · Acids, bases and salts

$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$

CHE-0045

RESPUESTA

FagusED · Chemistry
07 · Acids, bases and salts

The pH scale describes acidity and alkalinity. At 25 °C, pH below 7 is acidic, 7 neutral and above 7 alkaline. Universal indicator estimates pH across a colour range; litmus distinguishes acid from alkali; methyl orange and phenolphthalein give sharper end-points in titration. A lower pH means a higher H^+ concentration.

CHE-0044

RESPUESTA

FagusED · Chemistry
07 · Acids, bases and salts

Acids produce $\text{H}^+(\text{aq})$ ions; bases neutralise acids; alkalis are soluble bases that produce $\text{OH}^-(\text{aq})$. Acid + metal gives salt + hydrogen; acid + base gives salt + water; acid + carbonate gives salt + water + carbon dioxide. The salt name comes from the acid: hydrochloric forms chlorides, nitric forms nitrates, sulfuric forms sulfates.

CHE-0043

RESPUESTA

FagusED · Chemistry
07 · Acids, bases and salts

C) sulfates — The acid anion becomes sulfate in the salt.

CHE-0048

RESPUESTA

FagusED · Chemistry
07 · Acids, bases and salts

C) carbon dioxide — Carbonates release carbon dioxide with acids.

CHE-0047

RESPUESTA

FagusED · Chemistry
07 · Acids, bases and salts

A) H^+ — Acids release hydrogen ions in water.

CHE-0046

RESPUESTA

FagusED · Chemistry
08 · The Periodic Table

Lithium, sodium and potassium are soft, low-density metals forming +1 ions. They react with water to form metal hydroxide and hydrogen. Reactivity increases down the group because the outer electron is farther from the nucleus and more shielded, so attraction is weaker and electron loss is easier.

CHE-0051

RESPUESTA

FagusED · Chemistry
08 · The Periodic Table

Elements are arranged in increasing proton number. A period number gives the number of occupied electron shells; for main-group elements the group number relates to outer-shell electrons. Elements in the same group have similar chemical properties because they have the same number of valence electrons. Across a period, properties change from metallic to non-metallic.

CHE-0050

RESPUESTA

FagusED · Chemistry
07 · Acids, bases and salts

D) 13 — Values well above 7 are alkaline.

CHE-0049

RESPUESTA

FagusED · Chemistry
08 · The Periodic Table

State the key equation or relationship for group I: alkali metals.

CHE-0052

PREGUNTA

FagusED · Chemistry
08 · The Periodic Table

Elements are arranged by increasing
A) nucleon number
B) proton number
C) density
D) melting point

CHE-0053

PREGUNTA

FagusED · Chemistry
08 · The Periodic Table

An element with electron arrangement 2,8,2 is in
A) period 2 group II
B) period 3 group II
C) period 2 group III
D) period 3 group VIII

CHE-0054

PREGUNTA

FagusED · Chemistry
08 · The Periodic Table

Group I metals form ions with charge
A) 1+
B) 2+
C) 1-
D) 2-

CHE-0055

PREGUNTA

FagusED · Chemistry
08 · The Periodic Table

Group I reactivity down the group
A) decreases
B) increases
C) is constant
D) first rises then falls

CHE-0056

PREGUNTA

FagusED · Chemistry
09 · Metals

What should you recall about properties and uses?

CHE-0057

PREGUNTA

FagusED · Chemistry
09 · Metals

What should you recall about alloys?

CHE-0058

PREGUNTA

FagusED · Chemistry
09 · Metals

What exam pitfall should you avoid in rusting and corrosion?

CHE-0059

PREGUNTA

FagusED · Chemistry
09 · Metals

Why is copper used for electrical wiring?
A) low conductivity
B) conductivity and ductility
C) brittleness
D) high reactivity

CHE-0060

PREGUNTA

FagusED · Chemistry
09 · Metals

Why are alloys often harder than pure metals?
A) They contain molecules.
B) Different-sized atoms hinder layer sliding.
C) They have no electrons.
D) All atoms become identical.

CHE-0061

PREGUNTA

FagusED · Chemistry
09 · Metals

Which metal reacts most vigorously with cold water?
A) copper
B) iron
C) sodium
D) silver

CHE-0062

PREGUNTA

FagusED · Chemistry
09 · Metals

Which metal displaces copper from copper sulfate?
A) silver
B) gold
C) zinc
D) platinum

CHE-0063

PREGUNTA

FagusED · Chemistry
08 · The Periodic Table

B) period 3 group II — Three shells and two outer electrons.

CHE-0054

RESPUESTA

FagusED · Chemistry
08 · The Periodic Table

B) proton number — Proton number defines the periodic sequence.

CHE-0053

RESPUESTA

FagusED · Chemistry
08 · The Periodic Table



CHE-0052

RESPUESTA

FagusED · Chemistry
09 · Metals

Metals conduct heat and electricity, are malleable and ductile, and often have high density and melting point. Uses follow properties: copper wiring combines electrical conductivity and ductility; aluminium aircraft parts combine low density, strength and corrosion resistance. Chemical properties vary widely with reactivity.

CHE-0057

RESPUESTA

FagusED · Chemistry
08 · The Periodic Table

B) increases — The outer electron is lost more easily.

CHE-0056

RESPUESTA

FagusED · Chemistry
08 · The Periodic Table

A) 1+ — They lose one valence electron.

CHE-0055

RESPUESTA

FagusED · Chemistry
09 · Metals

B) conductivity and ductility — It carries charge and can be drawn into wires.

CHE-0060

RESPUESTA

FagusED · Chemistry
09 · Metals

A scratched zinc coating still protects iron sacrificially; a scratched tin coating may accelerate iron corrosion.

CHE-0059

RESPUESTA

FagusED · Chemistry
09 · Metals

An alloy is a mixture containing a metal and other elements. Different-sized atoms disrupt regular layers in the metal lattice, so layers slide less easily and the alloy is usually harder and stronger than the pure metal. Stainless steel contains iron with chromium and often nickel; brass contains copper and zinc.

CHE-0058

RESPUESTA

FagusED · Chemistry
09 · Metals

C) zinc — Zinc is more reactive than copper.

CHE-0063

RESPUESTA

FagusED · Chemistry
09 · Metals

C) sodium — Sodium is high in the reactivity series.

CHE-0062

RESPUESTA

FagusED · Chemistry
09 · Metals

B) Different-sized atoms hinder layer sliding. — Lattice distortion blocks easy movement of layers.

CHE-0061

RESPUESTA

FagusED · Chemistry
10 · Chemistry of the environment

What should you recall about water supply and treatment?

CHE-0064

PREGUNTA

FagusED · Chemistry
10 · Chemistry of the environment

What should you recall about fertilisers?

CHE-0065

PREGUNTA

FagusED · Chemistry
10 · Chemistry of the environment

State the key equation or relationship for air composition and pollutants.

CHE-0066

PREGUNTA

FagusED · Chemistry
10 · Chemistry of the environment

Potable water is water that is
A) pure H₂O only
B) safe to drink
C) always distilled
D) free of dissolved ions

CHE-0067

PREGUNTA

FagusED · Chemistry
10 · Chemistry of the environment

Anhydrous copper(II) sulfate changes from
A) blue to white
B) white to blue
C) pink to blue
D) green to red

CHE-0068

PREGUNTA

FagusED · Chemistry
10 · Chemistry of the environment

Which element in fertiliser is needed for amino acids?
A) nitrogen
B) argon
C) chlorine
D) neon

CHE-0069

PREGUNTA

FagusED · Chemistry
10 · Chemistry of the environment

What begins eutrophication after fertiliser runoff?
A) algal bloom
B) increased oxygen
C) fish reproduction
D) water distillation

CHE-0070

PREGUNTA

FagusED · Chemistry
11 · Organic chemistry

What should you recall about representing organic compounds?

CHE-0071

PREGUNTA

FagusED · Chemistry
11 · Organic chemistry

What should you recall about petroleum, fuels and cracking?

CHE-0072

PREGUNTA

FagusED · Chemistry
11 · Organic chemistry

What exam pitfall should you avoid in polymers?

CHE-0073

PREGUNTA

FagusED · Chemistry
11 · Organic chemistry

Successive members of a homologous series differ by
A) CH
B) CH₂
C) C₂H₂
D) OH

CHE-0074

PREGUNTA

FagusED · Chemistry
11 · Organic chemistry

Which is an alkane?
A) C₃H₆
B) C₄H₈
C) C₄H₁₀
D) C₂H₄

CHE-0075

PREGUNTA

FagusED · Chemistry
10 · Chemistry of the environment



CHE-0066

RESPUESTA

FagusED · Chemistry
10 · Chemistry of the environment

Plants require nitrogen for amino acids and proteins, phosphorus for roots and energy-transfer compounds, and potassium for enzyme function and healthy flowers and fruits. NPK fertilisers replace mineral ions removed from soil. Excess fertiliser can cause eutrophication: algal growth, light blocking, plant death, decomposition and oxygen depletion.

CHE-0065

RESPUESTA

FagusED · Chemistry
10 · Chemistry of the environment

Natural water may contain dissolved salts, microorganisms and suspended solids. Potable water is safe to drink, not chemically pure. Treatment commonly includes screening or sedimentation, filtration and chlorination. Distillation produces pure water but requires much energy. Tests for water include anhydrous copper(II) sulfate, white to blue, and cobalt(II) chloride paper, blue to pink.

CHE-0064

RESPUESTA

FagusED · Chemistry
10 · Chemistry of the environment

A) nitrogen — Nitrogen is required to make amino acids and proteins.

CHE-0069

RESPUESTA

FagusED · Chemistry
10 · Chemistry of the environment

B) white to blue — Hydration forms blue copper(II) sulfate.

CHE-0068

RESPUESTA

FagusED · Chemistry
10 · Chemistry of the environment

B) safe to drink — It is treated to be safe, though minerals can remain.

CHE-0067

RESPUESTA

FagusED · Chemistry
11 · Organic chemistry

Petroleum is a mixture of hydrocarbons separated by fractional distillation. Fractions condense over temperature ranges; smaller molecules have lower boiling points, are more volatile and ignite more easily. Cracking uses heat and a catalyst to convert long-chain alkanes into shorter alkanes and alkenes, matching demand and supplying monomers.

CHE-0072

RESPUESTA

FagusED · Chemistry
11 · Organic chemistry

A molecular formula gives numbers of each atom; a displayed formula shows every atom and bond; a structural formula shows connectivity in a compact form. Members of a homologous series share a functional group and general formula, have similar chemical properties, show gradual physical-property trends and differ successively by CH_2 .

CHE-0071

RESPUESTA

FagusED · Chemistry
10 · Chemistry of the environment

A) algal bloom — Nutrients stimulate rapid algal growth.

CHE-0070

RESPUESTA

FagusED · Chemistry
11 · Organic chemistry

C) C_4H_{10} — Alkanes follow $\text{C}_n\text{H}_{2n+2}$.

CHE-0075

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B) CH_2 — Each successive chain has one extra carbon and two hydrogens.

CHE-0074

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Keep continuation bonds through the brackets of a polymer repeat unit and include the subscript n.

CHE-0073

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Which property increases with hydrocarbon chain length?
A) volatility
B) ease of ignition
C) boiling point
D) flammability

CHE-0076

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Cracking produces
A) only long alkanes
B) shorter alkanes and alkenes
C) metals and oxygen
D) alcohols only

CHE-0077

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12 · Experimental techniques and chemical...

What should you recall about measurement and apparatus?

CHE-0078

PREGUNTA

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12 · Experimental techniques and chemical...

What should you recall about separation and purification?

CHE-0079

PREGUNTA

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12 · Experimental techniques and chemical...

State the key equation or relationship for chromatography.

CHE-0080

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12 · Experimental techniques and chemical...

Which apparatus transfers 25.0 cm³ most accurately?
A) beaker
B) pipette
C) conical flask
D) test tube

CHE-0081

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12 · Experimental techniques and chemical...

Which method separates an insoluble solid from a liquid?
A) filtration
B) distillation
C) chromatography
D) titration

CHE-0082

PREGUNTA

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12 · Experimental techniques and chemical...

Which method obtains pure water from salt solution?
A) filtration
B) simple distillation
C) crystallisation only
D) paper chromatography

CHE-0083

PREGUNTA

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12 · Experimental techniques and chemical...

Why is a chromatography baseline drawn in pencil?
A) Graphite dissolves readily.
B) Graphite does not dissolve in the solvent.
C) Ink is colourless.
D) Pencil increases R_f.

CHE-0084

PREGUNTA

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12 · Experimental techniques and chemical...

Choose apparatus according to precision and purpose. A burette measures variable accurate volumes; a volumetric pipette transfers one fixed accurate volume; a measuring cylinder is less precise; a gas syringe measures gas volume. Read liquid levels at eye level from the bottom of a concave meniscus and record burette readings to the appropriate precision.

CHE-0078

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B) shorter alkanes and alkenes — Large hydrocarbons split into smaller useful molecules.

CHE-0077

RESPUESTA

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C) boiling point — Stronger intermolecular forces raise boiling point.

CHE-0076

RESPUESTA

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12 · Experimental techniques and chemical...

B) pipette — A volumetric pipette transfers a fixed accurate volume.

CHE-0081

RESPUESTA

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12 · Experimental techniques and chemical...

$R_f = \text{distance moved by substance} / \text{distance moved by solvent front}$

CHE-0080

RESPUESTA

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12 · Experimental techniques and chemical...

Filtration separates an insoluble solid from a liquid. Crystallisation obtains a soluble solid by concentrating a solution and cooling it. Simple distillation obtains solvent or separates liquids with widely different boiling points; fractional distillation separates miscible liquids with closer boiling points. Purity is checked by a sharp melting or boiling point.

CHE-0079

RESPUESTA

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12 · Experimental techniques and chemical...

B) Graphite does not dissolve in the solvent. — It does not produce extra chromatogram spots.

CHE-0084

RESPUESTA

FagusED · Chemistry
12 · Experimental techniques and chemical...

B) simple distillation — Water vaporises and is condensed.

CHE-0083

RESPUESTA

FagusED · Chemistry
12 · Experimental techniques and chemical...

A) filtration — The solid remains in the filter.

CHE-0082

RESPUESTA